

The Evolution of Data Centres: From General-Purpose to Accelerated Computing

Data centres are being turned into modern-day factories for AI generation. It's no surprise, therefore, that traditional general-purpose computing is making way for accelerated computing.



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and application-specific integrated circuits (ASICs) to offload specific tasks from the CPU. These accelerators are designed to handle parallel processing tasks more efficiently, making them ideal for workloads that require massive computational power.

As artificial intelligence-related applications continue to grow, billions of dollars of capex are slated to be spent by data centre operators on accelerated computing hardware over the next decade.

AI requires three times the computation ability of the public cloud. It also needs more infrastructure and facilities. Global data centre capex is projected to grow 11% by 2027, reaching US\$ 400 billion, according to a market research firm. CPU refresh cycles, accelerated computing and edge computing are the main elements driving these investments.

By incorporating a mix of accelerators alongside traditional CPUs, data centres can optimise their infrastructure for different types of tasks and applications.

Role of GPUs in accelerated computing

GPUs, or graphics processing units, play a crucial role in accelerated computing by leveraging their parallel processing power to perform complex calculations and tasks much faster

Data centres have been at the forefront of tech innovation and transformation for quite some time now. One of the most significant shifts they have experienced in recent years has been the move from general-purpose computing to accelerated computing. This shift has been driven by the demand for faster processing speeds, improved efficiency, and the need to handle complex workloads such as artificial intelligence, machine learning, and big data analytics.

General-purpose computing, which relies on traditional central processing units (CPUs), has long been the standard in data centres. However, as the volume and complexity of data continue to grow exponentially, CPUs have struggled to keep up with the processing demands. This has led to a search for alternative solutions that can offer higher performance and efficiency.

Accelerated computing leverages specialised hardware such as graphics processing units (GPUs), field-programmable gate arrays (FPGAs),